# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 30%*

Annexure A

**CONCEPT MAPPING**

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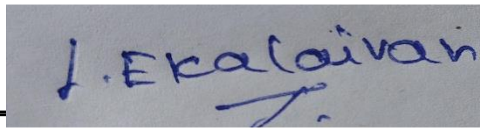
***Code of Conduct:***

*I EKALAIVAN L(192524450)__~~(Name / Reg No) certify that this submission is my~~*
*original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



Annexure A

## CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______→ Transport → Network → ______→ Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

**RUBRICS FOR CALCULATION**

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Concept Identification** | 20% | Accurately identifies all relevant OSI layers, concepts, and relationships | Identifies most concepts with minor omissions | Identifies limited concepts; some inaccuracies | Unable to identify key concepts correctly |
| **Concept Mapping Structure** | 25% | Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections | Concept map is mostly clear with minor structural issues | Concept map lacks clarity, organization, or proper flow | No clear structure or incorrect mapping |
| **Linking & Relationships** | 20% | Clearly establishes correct relationships between layers, functions, and processes | Relationships are mostly correct with minor errors | Limited or partially correct relationships | Incorrect or missing relationships |
| **Application & Analysis** | 20% | Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning | The application is correct but lacks depth in analysis | Basic application with weak explanation | Unable to apply concepts correctly |
| **Explanation & Clarity** | 15% | Provides a clear, concise, and well-structured explanation supporting the concept map | Explanation is understandable with minor gaps | Explanation is unclear or incomplete | No clear or meaningful explanation |

## ANSWERS:

**1. Missing OSI Layers**

**Concept Map**

**Application**
↓
**Presentation**
↓
**Session**
↓
**Transport**
↓
**Network**
↓
**Data Link**
↓
**Physical**

**Explanation**

**The missing layers are Session Layer and Data Link Layer.**

**The Session Layer establishes, manages, synchronizes, and terminates communication sessions between two devices. It inserts synchronization points (checkpoints) so that communication can continue from the last checkpoint if a failure occurs.**

**The Data Link Layer provides node-to-node communication by converting packets into frames. It performs framing, physical addressing (MAC address), error detection using CRC, and flow control to ensure reliable transmission over the physical medium.**

**Analysis**

**The Session Layer ensures continuous communication by maintaining active sessions, while the Data Link Layer guarantees error-free frame transmission between neighboring devices. Together with the other OSI layers, they provide reliable and efficient end-to-end communication.**

**2. Data Units and OSI Layers**

**Concept Map**
**Application**
**Presentation**
**Session**
↓
**Transport → Segment**
↓
**Network → Packet**
↓
**Data Link → Frame**
↓
**Physical → Bits**

**Explanation**

**When user data is transmitted, each OSI layer converts it into a different Protocol Data Unit (PDU).**

**Transport Layer → Segment**
**Network Layer → Packet**
**Data Link Layer → Frame**
**Physical Layer → Bits**

**During transmission, the sender performs encapsulation, where each layer adds its own header. At the receiver, decapsulation removes these headers in reverse order to recover the original data.**

**Analysis**

**The transformation of data units ensures that addressing, routing, error detection, and physical transmission are handled correctly. This layered approach simplifies communication and improves reliability across different network technologies.**

**3. MAC Address and IP Address**

**Concept Map**
**Data Link Layer**
↓
**MAC Address**
↓
**Local Delivery**

**Network Layer**
↓
**IP Address**
↓
**Global Delivery**
**Explanation**

**The MAC Address is a unique hardware address used for communication within the same Local Area Network (LAN). Switches use MAC addresses to deliver frames between devices.**

**The IP Address is a logical address used by routers to forward packets between different networks across the Internet.**

**Analysis**

**When a user sends data to another computer on a different network, the IP address identifies the destination network, while the MAC address identifies the next local device. The cooperation of these two addressing methods enables complete end-to-end communication.**

## 4. Error Control

**Concept Map**
Error Control

Data Link Layer   Transport Layer

Error Detection   Error Correction

CRC, Checksum   ACK, Retransmission

**Explanation**

The Data Link Layer detects transmission errors using techniques such as Cyclic Redundancy Check (CRC) and checksum before forwarding frames.

The Transport Layer ensures reliable communication using acknowledgements (ACKs), sequence numbers, flow control, and retransmission of lost segments.

**Analysis**

Both layers work together to improve reliability. The Data Link Layer detects errors on individual links, while the Transport Layer guarantees complete and correct delivery between the source and destination. This minimizes data loss and improves network performance.

## 5. Web Browsing Using OSI Layers

**Concept Map**
Application → HTTP/HTTPS
↓
Presentation → SSL/TLS Encryption
↓
Session → Session Management
↓
Transport → TCP
↓
Network → IP Routing
↓
Data Link → Ethernet/Wi-Fi
↓
Physical → Cable/Fiber/Wireless Signals

**Explanation**

When a user opens a website, the Application Layer uses HTTP/HTTPS to request the webpage. The Presentation Layer encrypts and decrypts data using SSL/TLS. The Session Layer establishes and maintains communication sessions. The Transport Layer uses TCP to ensure reliable delivery through acknowledgements and retransmissions. The Network Layer routes packets using IP addresses. The Data Link Layer delivers frames within the local network, while the Physical Layer transmits bits through cables or wireless signals.

**Analysis**

Each layer depends on the services of the layer below it. If any layer fails, communication is interrupted. For example:

Physical Layer failure → No signal transmission.
Data Link Layer failure → Frames cannot be delivered.
Network Layer failure → Packets cannot reach the destination.
Transport Layer failure → Reliable communication is lost.
Application Layer failure → The webpage cannot be displayed.

The layered architecture simplifies troubleshooting and ensures efficient web communication.

6. Network Troubleshooting

Concept Map
Physical ✓
↓
Data Link ✓
↓
Network ✕
↓
Transport ✕
↓
Application ✕
Explanation

The concept map shows that the Physical Layer and Data Link Layer are functioning correctly because cables, signals, and local frame delivery are working. However, the Network Layer has failed, preventing packets from being routed to the destination. As a result, the Transport Layer cannot establish reliable communication, and the Application Layer cannot provide services to the user.

Possible causes of the Network Layer failure include:

Incorrect IP address configuration
Router failure
Routing table errors
IP address conflicts
Analysis

The concept map helps administrators quickly identify the faulty layer. Since the lower layers are working properly, troubleshooting should begin at the Network Layer by checking IP configuration, routing tables, gateway settings, and router connectivity. This systematic approach reduces troubleshooting time and improves network reliability.